# Medhansh Garg

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## Education

- **University of Illinois, Urbana-Champaign** Aug. 2023 - May 2027 (Expected)
  - B.S. in Computer Engineering - Computer Systems Engineering, Analog Signal Processing, Digital Forensics
- **NYU MicroBachelors in Cybersecurity** 2023
  - 1 year program auth/access control, network security, penetration testing, recieved A grade

## Skills

- **Languages**: TypeScript, Python, C, C++, Java
- **Frameworks**: Next.js, React, Flask, Spring Boot
- **Security & Reverse Engineering**: Burp Suite, Frida, JADX, Wireshark, Nmap
- **Infra & Embedded**: Docker, Kubernetes, Azure, Firebase, Supabase, Vercel, ESP32, Raspberry Pi

## Experience

- **CS 124 Honors** | *Lead Web Developer* Aug. 2025 - Present
  - Implemented UIUC SSO login utilizing **CILogon** and Banner API **role mapping**, streamlining access and enhancing platform security
  - Leading **11-person dev team** serving **200+ students**; run weekly meetings, delegate tasks, review pull requests
- **Care Health Insurance** | *Cybersecurity Intern* Jun. 2025 - Aug. 2025
  - Discovered **60+ unauthenticated internet-exposed services** via external recon; reported findings to drive remediation, securing confidential client data
  - Reverse-engineered **WAF request-signing**; built **replay attack** PoC in **Burp Suite**
  - Performed Android security testing using **Frida** + **JADX**; Bypassed **SSL pinning** and **root-detection**
- **IoT++** | *Software Engineering Intern* Jun. 2024 - Aug. 2024
  - Built real-time fire/smoke detection pipeline using **YOLOv8** for deployment in manufacturing plant camera feeds
  - Improved detection accuracy by **62%** over baseline and **13x** throughput via **pruning** + **tiling**
  - Deployed **NPU-optimized** inference on Orange Pi using quantization + **GStreamer**
  - Containerized inference with **Docker** + **Kubernetes** on **Azure**

## Leadership & Involvement

- **Illinois MicroTech** | *HackerFab Member* | *EOH Project Lead* Oct. 2025 - Present
  - *HackerFab:* Engineering **3-axis stepper-driven** stage with **sub-micron positioning** for wafer photolithography
  - *EOH 2026:* Led Engineering Open House showcase for **20,000+ attendees** as project lead
- **Sigpwny, Cybersecurity RSO** | *Badge Developer* Jan. 2025 - Sep. 2025
  - Engineered and manufactured a DEFCON-style PCB badge (**250 units**) for a UIUC-hosted CTF even
  - Designed **ESP32 hardware** + **MicroPython firmware** with SAO modular expansion
  - Built **frame-buffer LCD display stack** and badge **UI/UX**

## Projects

- **AlberFlowy** https://github.com/HackOverflow404/AlberFlowy
  - Reverse-engineered WorkFlowy private API for full tree CRUD via **Node.js CLI**
  - Built **C++** Albert plugin with **caching** + **async sync** + **Google OAuth2 automation**
- **RemoteCam** https://github.com/HackOverflow404/RemoteWebcam
  - Built cross-platform **WebRTC** webcam streaming system with **Firebase** signaling
  - **QtPython** Linux client + **Next.js** mobile **PWA** for real-time camera sharing
- **Spotify Wall Art** https://github.com/HackOverflow404/SpotifyWebController
  - Repurposed laptop **LCD** with **Raspberry Pi** to build live album-art display
  - Integrated **Spotify API** for real-time track data, lyrics, and animations

## Publications

- **Biometric authentication and cybersecurity in the digital age** - *Publication*: MedCrave
  - Compared **biometric vs traditional** authentication; proposed layered auth model. DOI: 10.15406/bbij.2025.14.00437
- **Evaluation of Leet Speak on Password Strength and Security** - *Publication*: IJSRST
  - Authored paper on Leet Speak impact on password strength (**834K+ samples**). DOI: 10.32628/IJSRST229567